

Homework #1

Due time: 2:20 pm EST, Monday, Jan 29

1. Thinking purely in geometric terms, estimate the diameter of the smallest printed dot that the eye can discern if the page on which the dot is printed is 0.4 m away from the eyes. (25pts)

Assume for simplicity that

- the distance between the center of the lens and the retina along the visual axis is 14mm;
- the visual system ceases to detect the dot when the image of the dot on the fovea becomes smaller than the diameter of one receptor (cone) in that area of the retina; and
- the fovea can be modeled as a square array of dimensions 1.5 mm*1.5 mm, and that the cones (about 337,000 in total) and spaces between the cones are distributed uniformly throughout this array.

Solution:

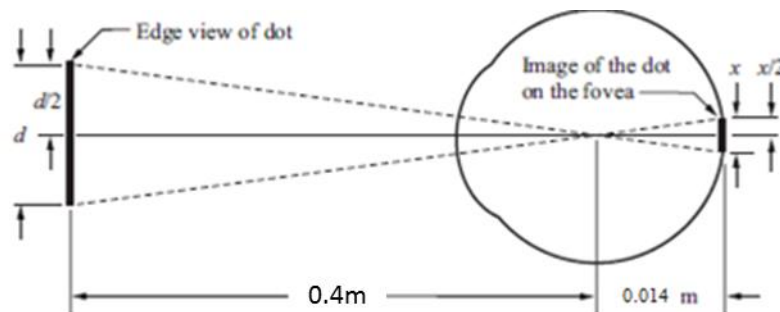


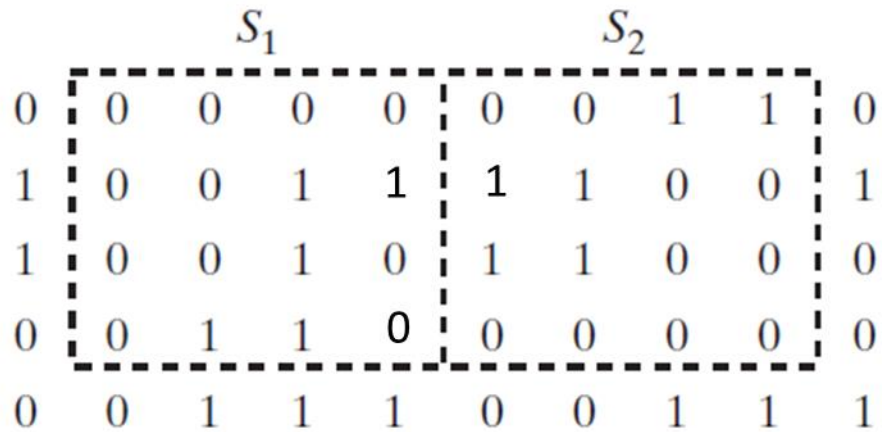
Figure 1

The diameter, x , of the retinal image corresponding to the dot is obtained from similar triangles, as shown in below. That is,

$$\frac{d}{0.4} = \frac{x}{0.014}$$

which gives $x = 0.035d$. From the discussion in Section 2.1.1, we can think of the fovea as a square sensor array having on the order of 337,000 elements, which translates into an array of size 580×580 elements. Assuming equal spacing between elements, this gives 580 elements and 579 spaces on a line 1.5 mm long. The size of each element and each space is then $s = [(1.5\text{mm})/1,159] = 1.3 \times 10^{-6}$ m. If the size (on the fovea) of the imaged dot is less than the size of a single resolution element, we assume that the dot will be invisible to the eye. In other words, the eye will not detect a dot if its diameter, d , is such that $0.035(d) < 1.3 \times 10^{-6}$ m, or $d < 37.14 \times 10^{-6}$ m.

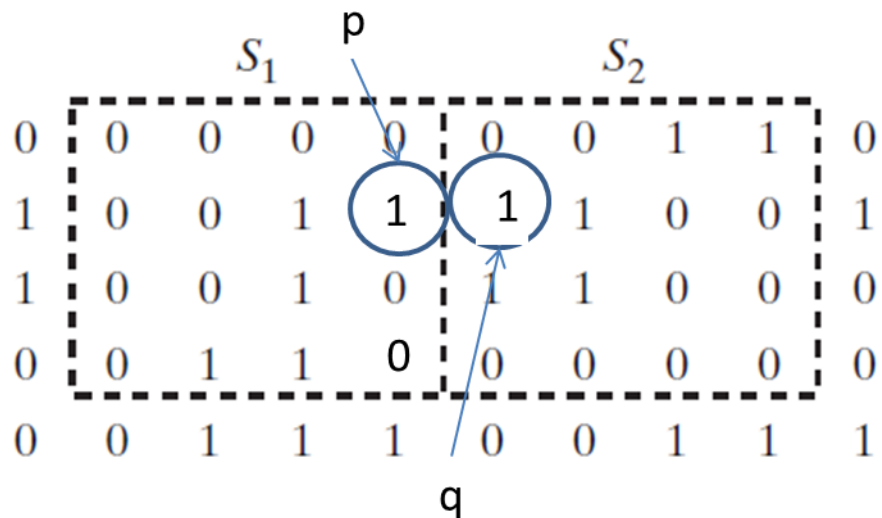
2. Consider the two image subsets, S_1 and S_2 , shown in the following figure. For $V=\{1\}$, determine whether these two subsets are (a) 4-adjacent, (b) 8-adjacent, or (c) m-adjacent. (25pts)



Solution:

Let p and q be the two pixels in S_1 and S_2 , as shown below. Then,

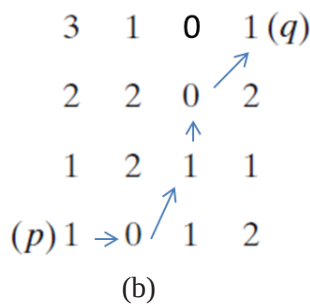
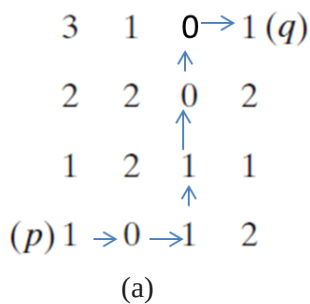
- (a) S_1 and S_2 are not 4-adjacent because S_2 *itself* is not a connected set in terms of 4-adjacency.
- (b) S_1 and S_2 are 8-adjacent because both S_1 and S_2 are connected sets in terms of 8-adjacency and $S_1 \cup S_2$ is a connected set in terms of 8-adjacency.
- (c) S_1 and S_2 are m -adjacent because both S_1 and S_2 are connected sets in terms of m -adjacency and $S_1 \cup S_2$ is a connected set in terms of m -adjacency.



3. Consider the image segment shown below. (25pts)
- (a) Let $V=\{0, 1\}$ and compute the lengths of the shortest 4-, 8-, and m -path between p and q . If a particular path does not exist between these two points, explain why.
- (b) Repeat for $V=\{1, 2\}$.

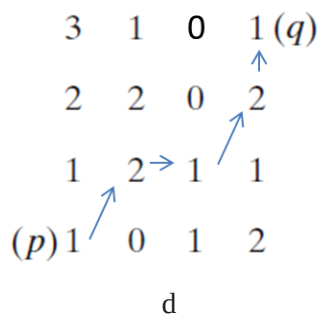
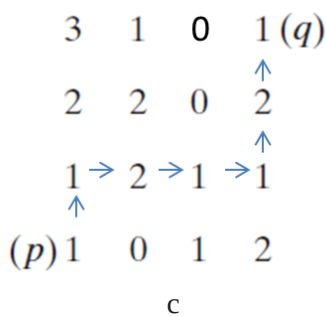
	3	1	0	1 (q)
	2	2	0	2
	1	2	1	1
(p)	1	0	1	2

(a) When $V = \{0,1\}$, the shortest 4-path is shown in Figure a, with length 6. The shortest 8-path is shown in Figure b, with length 4. The shortest m - path is the same as the shortest 4-path with length 6 (shown in Figure a).

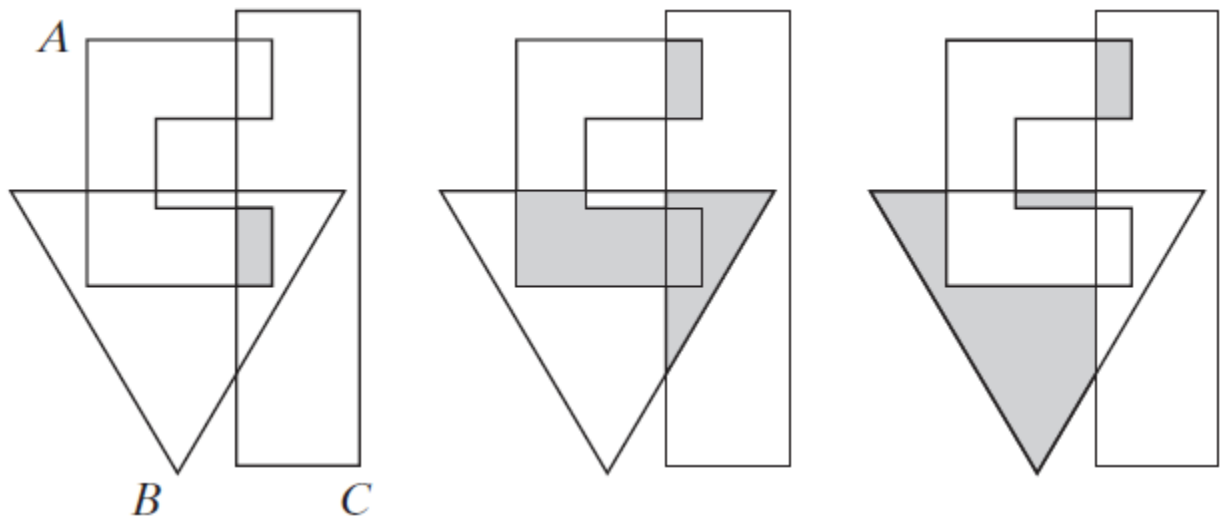


(a) The unique shortest 4-path when $V = \{1,2\}$ is shown in Figure c; its length is 6.

The unique shortest 8-path is shown in Figure d; its length is 4. The shortest m - path is the same as the shortest 4-path with length 6 (shown in Figure c).



4. Give expressions for the sets shown shaded in the following figure in terms of sets A, B, and C. The shaded areas in each figure constitute one set, so give one expression for each of the three figures. (25pts)



Solution:

The answers are, from left to right,

$$A \cap B \cap C \quad (1)$$

$$(A \cap C) \cup (A \cap B) \cup (B \cap C) \quad (2)$$

$$[(B - A) - C] \cup [(A \cap C) - (A \cap B \cap C)] \quad (3)$$

$$\text{or } [B \cap (A \cup C)^c] \cup [A \cap (C - B)] \quad (3)$$

$$\text{or } [B - (A \cup C)] \cup [(A \cap C) - (A \cap B \cap C)] \quad (3)$$